Decision Science
Problem Taxonomy Guide

Statistics Method Family — Working Session Scope

DDSI Decision Design Assistant — Bootcamp Reference

Purpose

This document defines the standard problem categories used by the Decision Science team to classify incoming analytical requests. When a stakeholder or product lead describes a business problem, the first step is to map it to one of the categories below. This ensures alignment on what type of problem is being solved before any modeling or experimentation begins.

How to Use This Guide

1. Listen to the stakeholder's description of their problem or question.
2. Identify which category below best matches the **decision** the stakeholder is trying to make.
3. Confirm the classification with the stakeholder using plain language — avoid jargon.
4. **Operationalize the language** — restate any vague or ambiguous terms with precise definitions. If they say "young families," define what that means. Never leave business terms undefined.
5. **Check for compound problems** — stakeholders often bundle multiple problems together. Decompose compound requests into components and clarify which is primary.
6. Document the agreed-upon problem category, decision intent, operationalized definitions, and success criteria before proceeding to feasibility assessment.

*For this working session, all categories fall within the **Statistics** method family. See the appendix for the expansion roadmap to other method families.*

1. Hypothesis Testing

Definition: Determining whether there is statistically significant evidence to support or reject a specific claim about a population, process, or treatment effect.

Decision Intent: The stakeholder has a specific belief or claim and needs evidence to confirm or refute it before acting.

Key Indicators in Stakeholder Language:

- "Is there a real difference between...?"
- "Did [change] actually make a difference?"
- "Are these two groups really different, or is it just noise?"
- "Can we prove that...?"

Typical Success Criteria:

- Clear null and alternative hypotheses defined upfront
- Appropriate significance level agreed upon (e.g., $\alpha = 0.05$)
- Sufficient sample size to detect meaningful effects
- Results communicated with p-values and confidence intervals

Example Scenarios:

- *We changed the layout of the merchandise store in Tomorrowland. Did average transaction value actually increase?*
- *Our new onboarding process is supposed to reduce employee ramp-up time. Did it work?*

2. Regression & Correlation Analysis

Definition: Quantifying the relationship between a dependent variable and one or more independent variables to understand drivers, estimate effects, or make predictions.

Decision Intent: The stakeholder wants to understand what factors drive an outcome, how strong those relationships are, or what to expect given certain inputs.

Key Indicators in Stakeholder Language:

- "What drives...?"
- "How does [X] affect [Y]?"
- "If we change [X], what happens to [Y]?"
- "What are the most important factors?"

Typical Success Criteria:

- Model explains a meaningful portion of variance
- Key drivers identified with coefficient estimates and CIs
- Assumptions validated (linearity, normality, homoscedasticity)
- Results interpretable by non-technical stakeholders

Example Scenarios:

- *What factors most influence guest satisfaction scores at our resort hotels?*
- *How much does weather affect daily park attendance?*

3. Time Series Analysis & Forecasting

Definition: Analyzing data collected over time to identify patterns (trends, seasonality, cycles) and generate forecasts for future time periods.

Decision Intent: The stakeholder needs to understand temporal patterns or predict future values to support planning decisions.

Key Indicators in Stakeholder Language:

- "What's the trend in...?"
- "How many [X] should we expect next [period]?"
- "Is this seasonal?"
- "Are things getting better or worse over time?"

Typical Success Criteria:

- Forecast accuracy within acceptable error band (e.g., MAPE < 10%)
- Forecasts at required granularity (daily, weekly, monthly)
- Confidence intervals alongside point estimates
- Seasonal and trend components clearly decomposed

Example Scenarios:

- *Predict park attendance for Spring Break 2027 to plan staffing.*
- *How does merchandise revenue typically behave across the fiscal year?*

4. Estimation & Confidence Intervals

Definition: Estimating an unknown population parameter (mean, proportion, variance) from sample data and quantifying the uncertainty of that estimate.

Decision Intent: The stakeholder needs a reliable estimate of a quantity with a clear understanding of how precise it is.

Key Indicators in Stakeholder Language:

- "What's the average...?"
- "What percentage of...?"
- "How confident are we in this number?"
- "Give me a ballpark with a margin of error."

Typical Success Criteria:

- Point estimate with CI at agreed confidence level
- Sample size sufficient for desired margin of error
- Assumptions about the population validated
- Results in business-friendly language

Example Scenarios:

- *What's the average per-guest spending at Magic Kingdom, and how confident are we?*
- *What proportion of guests use mobile ordering?*

5. Distribution Fitting & Descriptive Analytics

Definition: Characterizing the shape, spread, and central tendency of data to understand its underlying distribution and summarize key properties.

Decision Intent: The stakeholder wants to understand the fundamental nature of their data — what's typical, what's unusual, what patterns exist.

Key Indicators in Stakeholder Language:

- "What does the data look like?"
- "What's normal for...?"
- "Are there outliers?"
- "Help me understand this data."

Typical Success Criteria:

- Clear summary statistics (mean, median, SD, percentiles)
- Distribution shape identified
- Outliers identified and contextualized
- Visualizations that tell the story

Example Scenarios:

- *What does the distribution of wait times look like across all attractions?*
- *Are there outliers in our F&B; transaction data that might be errors?*

6. Comparative Analysis (Multi-Group)

Definition: Comparing metrics, distributions, or outcomes across three or more groups or conditions to identify meaningful differences.

Decision Intent: The stakeholder wants to know whether different groups produce different results, and which ones stand out.

Key Indicators in Stakeholder Language:

- "How does [X] compare across...?"
- "Which [location/segment] performs best?"
- "Are there differences between...?"
- "Is one approach better than the others?"

Typical Success Criteria:

- Appropriate test selected (ANOVA, Kruskal-Wallis, chi-square)
- Post-hoc tests identify which groups differ
- Effect sizes reported alongside significance
- Multiple comparison corrections applied

Example Scenarios:

- *Compare guest satisfaction across all four WDW parks — are there real differences?*
- *Is the demographic mix different across weekdays vs. weekends vs. holidays?*

7. Sampling & Survey Design

Definition: Designing data collection strategies that produce representative, unbiased samples and determining appropriate sample sizes.

Decision Intent: The stakeholder needs to collect new data and wants to ensure the approach produces reliable results.

Key Indicators in Stakeholder Language:

- "How many responses do we need?"
- "Is our sample big enough?"
- "How should we select who to survey?"
- "Will this sample be representative?"

Typical Success Criteria:

- Sample size calculated for desired margin of error and confidence
- Sampling strategy minimizes selection bias
- Data collection plan is operationally feasible

Example Scenarios:

- *We want to survey guests about a new dining experience. How many responses do we need?*
- *We need to audit merchandise inventory across 200 locations. Can we sample?*

8. Bayesian & Probabilistic Estimation

Definition: Incorporating prior knowledge with observed data to update probability estimates, particularly useful when data is limited or expert judgment is valuable.

Decision Intent: The stakeholder has prior context and wants to combine it with new data to make a more informed estimate with limited data.

Key Indicators in Stakeholder Language:

- "We don't have much data, but we have experience..."
- "How should we update our expectations?"
- "What's the probability that...?"
- "We need to decide with limited data."

Typical Success Criteria:

- Prior distribution justified and documented
- Posterior distribution clearly communicated
- Results as probability statements stakeholders can act on
- Sensitivity analysis on priors

Example Scenarios:

- *We're launching a new attraction with no historical data. What's the probability daily ridership exceeds 10,000?*
- *We have 6 months of data on a new loyalty program. Is it too early to tell if it's working?*

Quick Classification Reference

If the problem involves...	Category
Testing a specific claim	Hypothesis Testing
Understanding what drives an outcome	Regression & Correlation
Predicting future values over time	Time Series Analysis
Estimating a quantity with uncertainty	Estimation & CIs
Understanding data shape and patterns	Distribution Fitting
Comparing groups or conditions	Comparative Analysis
Designing data collection	Sampling & Survey Design
Limited data with strong prior knowledge	Bayesian Estimation

Detecting Compound ("Trojan Horse") Problems

Stakeholders often present what appears to be a single problem that actually contains multiple distinct analytical problems. Warning signs:

- **Multiple verbs:** "predict demand AND optimize pricing" — two separate problems
- **Expanding scope mid-conversation:** "Oh, and we'd also need to know..."
- **Different data needs for different parts:** one part needs transactions, another needs surveys
- **Mix of descriptive and predictive goals:** "Tell me what's happening AND what will happen"

When you detect a compound problem:

1. List each distinct sub-problem
2. Classify each one separately
3. Identify the **primary** problem (most directly supports the decision)
4. Note dependencies between sub-problems
5. Recommend sequential, parallel, or separate workstreams

Example:

Stakeholder says: "I need to figure out the best pricing to attract young families to special events at the parks."

This contains: (1) Descriptive Analytics — defining "young families" in data; (2) Regression — understanding what drives this segment's attendance; (3) Pricing optimization — outside statistics scope. Primary for statistics: #2, with #1 as prerequisite.

Operationalizing Business Language

Convert vague business terms into precise, measurable definitions. Always ask: "When you say [term], can you help me define exactly what you mean?"

Vague Term	Questions to Ask	Example Definition
Young families	Age range? Children's ages?	Parents 25-40, child under 12
Peak season	Which dates? Based on what?	Weeks with attendance > 85th pctl
High spenders	Threshold? Per visit or annual?	In-park spend > 90th pctl per visit
Regular guests	Frequency? Time period?	3+ visits in trailing 12 months
Performing well	What metric? vs. what baseline?	Revenue growth > 5% YoY, trailing 4Q
Special events	Which qualify? Ticketed only?	Separately ticketed after-hours events

Appendix: Broad Method Family Taxonomy (Future Scope)

The statistics-focused categories in this guide are a subset of a broader taxonomy the team plans to expand into over time:

Method Family	Description	Status
Statistics	Hypothesis testing, regression, estimation, time series, descriptive	**Active — this guide**
Optimization	Linear programming, integer programming, constraint satisfaction	Future expansion
Machine Learning	Classification, clustering, ensemble methods, deep learning	Future expansion
Causal Modeling	A/B testing design, diff-in-diff, propensity scoring	Future expansion
Simulation	Monte Carlo, agent-based modeling, discrete event simulation	Future expansion
Decision Analysis	Decision trees, multi-criteria decision analysis, risk scoring	Future expansion